

Model Lab

Every prior, same slate — 2026-09-18

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

36,388

Held-Out Scores

Fitted on plate appearances before 2026-08-12, scored on the 36,388 that came after (actual HR rate 2.98%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5713	1.38×	0.02886	0.13323	0.00338
KNN comps Prior is the pooled rate of the nearest comparables.	0.5878	1.54×	0.02885	0.13287	0.00489
SVR Support-vector regression on the same feature matrix.	0.5766	1.40×	0.02893	0.13453	0.00835
Random forest Forest regression; a learned similarity metric.	0.5796	1.45×	0.02890	0.13392	0.00718
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5878	1.54×	0.02885	0.13287	0.00489
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5873	1.49×	0.02885	0.13293	0.00503
Linear regression Ordinary least squares on the same features.	0.5849	1.42×	0.02887	0.13356	0.00625
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5870	1.51×	0.02888	0.13327	0.00704
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5865	1.51×	0.02887	0.13319	0.00639
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5770	1.44×	0.02894	0.13438	0.00831

Blend weights (forest variant): KNN 0.90, SVR 0.00, RF 0.10

Shrinkage K per prior: league 247, knn 247, svr 247, rf 247, core 247, ensemble 247, linear 247, logistic 247, nn 247, xgb 247 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,888 PA, actual HR rate 2.55%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5127	1.01×	0.02483	0.11894	0.00624
KNN comps Prior is the pooled rate of the nearest comparables.	0.5674	1.31×	0.02488	0.12024	0.01260
SVR Support-vector regression on the same feature matrix.	0.5438	1.21×	0.02530	0.13067	0.02325
Random forest Forest regression; a learned similarity metric.	0.5533	1.21×	0.02514	0.12670	0.01907
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5674	1.31×	0.02488	0.12024	0.01260
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5672	1.31×	0.02490	0.12062	0.01299
Linear regression Ordinary least squares on the same features.	0.5670	1.31×	0.02506	0.12589	0.01743
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5700	1.21×	0.02500	0.12219	0.01609
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5742	1.51×	0.02495	0.12165	0.01528
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5459	1.11×	0.02541	0.12963	0.02509

Slate Agreement

Flat league · 3/6 shared 1/6 hit

Pete Crow-Armstrong

Elly De La Cruz

Nolan Arenado

Nick Sogard

Jeremy Peña

Jahmai Jones ✓

SVR · 4/6 shared 1/6 hit

Pete Crow-Armstrong

Elly De La Cruz

Junior Caminero

Yordan Alvarez

Jahmai Jones ✓

Emmanuel Rodriguez

Core (KNN+SVR) · 6/6 shared 2/6 hit

Pete Crow-Armstrong

Jahmai Jones ✓

Lazaro Montes ✓

Michael Conforto

Elly De La Cruz

Yordan Alvarez

Linear regression · 5/6 shared 2/6 hit

Pete Crow-Armstrong

Lazaro Montes ✓

Emmanuel Rodriguez

Jahmai Jones ✓

Elly De La Cruz

Yordan Alvarez

Neural net (torch) · 5/6 shared 2/6 hit

KNN comps · 6/6 shared 2/6 hit

19.9% Pete Crow-Armstrong

15.7% Jahmai Jones ✓

14.6% Lazaro Montes ✓

14.5% Michael Conforto

14.5% Elly De La Cruz

14.4% Yordan Alvarez

Random forest · 5/6 shared 2/6 hit

22.9% Pete Crow-Armstrong

18.3% Lazaro Montes ✓

17.5% Emmanuel Rodriguez

17.5% Jahmai Jones ✓

17.3% Elly De La Cruz

17.0% Yordan Alvarez

Forest blend · 6/6 shared 2/6 hit

21.9% Pete Crow-Armstrong

18.5% Jahmai Jones ✓

17.8% Lazaro Montes ✓

16.9% Michael Conforto

16.7% Elly De La Cruz

16.1% Yordan Alvarez

Logistic regression · 3/6 shared 1/6 hit

21.9% Lazaro Montes ✓

21.1% Emmanuel Rodriguez

19.1% Pete Crow-Armstrong

17.1% Yordan Alvarez

16.8% Kyle Schwarber

16.5% Ben Ross

21.9%

18.5%

17.8%

16.9%

16.7%

16.1%

22.6%

18.5%

17.6%

17.2%

17.2%

16.9%

21.9%

18.3%

17.9%

16.8%

16.8%

16.2%

25.5%

25.5%

23.4%

18.0%

17.5%

17.4%

Yordan Alvarez

Ethan Salas

17.2%

Hitter by Hitter

[illegible]

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. Brandon Lowe Pittsburgh Pirates · vs RHP · 31 HR / 620 PA	14.1%	15.8%	16.3%	16.3%	15.8%	15.9%	15.9%	16.8%	16.3%
9. Emmanuel Rodriguez Minnesota Twins · vs LHP · 1 HR / 24 PA	10.4%	15.8%	17.0%	17.6%	15.8%	16.0%	19.1%	25.5%	17.3%
10. Kazuma Okamoto Toronto Blue Jays · vs RHP · 33 HR / 600 PA	13.8%	15.8%	16.8%	15.9%	15.8%	15.8%	15.6%	16.5%	16.2%
11. Ben Rice New York Yankees · vs RHP · 38 HR / 628 PA	13.6%	15.7%	16.8%	16.3%	15.7%	15.7%	15.6%	16.0%	16.0%
12. Carter Jensen Kansas City Royals · vs RHP · 24 HR / 571 PA	13.5%	15.6%	15.7%	15.1%	15.6%	15.6%	14.6%	15.2%	15.4%
13. Ian Happ Chicago Cubs · vs RHP · 23 HR / 602 PA	13.2%	15.6%	15.1%	15.4%	15.6%	15.5%	15.2%	15.4%	15.3%
14. Corey Seager Texas Rangers · vs RHP · 18 HR / 398 PA	12.3%	15.5%	15.0%	15.2%	15.5%	15.4%	15.1%	15.4%	15.5%
15. Leo Bernal St. Louis Cardinals · vs RHP · 3 HR / 65 PA	11.5%	15.4%	13.6%	14.7%	15.4%	15.3%	14.5%	14.0%	14.4%
16. JJ Bleday Cincinnati Reds · vs RHP · 22 HR / 512 PA	13.4%	15.3%	15.6%	14.9%	15.3%	15.3%	14.8%	15.0%	15.0%
17. Matt Olson Atlanta Braves · vs RHP · 39 HR / 662 PA	13.2%	15.3%	16.4%	16.0%	15.3%	15.3%	15.3%	16.5%	16.0%
18. Heriberto Hernández Miami Marlins · vs RHP · 24 HR / 465 PA	13.0%	15.2%	15.1%	15.2%	15.2%	15.2%	15.1%	15.4%	15.4%
19. Munetaka Murakami Chicago White Sox · vs RHP · 32 HR / 512 PA	12.8%	15.1%	15.4%	15.6%	15.1%	15.2%	14.9%	15.2%	15.4%
20. Coby Mayo Baltimore Orioles · vs LHP · 24 HR / 420 PA	12.6%	15.0%	16.0%	15.5%	15.0%	15.1%	15.2%	16.5%	15.7%
21. Mark Vientos New York Mets · vs LHP · 13 HR / 273 PA	11.9%	15.0%	13.7%	14.6%	15.0%	15.0%	14.9%	15.5%	15.2%
22. Yohandy Morales	11.0%	15.0%	14.8%	14.8%	15.0%	15.0%	14.0%	14.4%	15.1%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Kody Clemens Minnesota Twins · vs LHP · 26 HR / 553 PA	11.7%	14.3%	14.2%	14.4%	14.3%	14.3%	14.0%	14.8%	14.4%
38. Eugenio Suárez Cincinnati Reds · vs RHP · 24 HR / 470 PA	13.1%	14.2%	14.4%	14.5%	14.2%	14.3%	14.4%	14.5%	14.3%
39. Ethan Salas San Diego Padres · vs RHP · 2 HR / 21 PA	11.0%	14.2%	16.3%	16.1%	14.2%	14.4%	15.4%	14.5%	15.0%
40. Cal Raleigh Seattle Mariners · vs RHP · 22 HR / 495 PA	11.8%	14.2%	13.1%	13.7%	14.2%	14.2%	14.5%	15.4%	14.2%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.